

7 The photograph shows a yak.



(Source: Zoonar GmbH / Alamy Stock Photo)

Yaks are cattle that live at high altitudes, such as in the Himalayas.

They are well-adapted for the low temperatures and low levels of oxygen in the air.

These adaptations include:

- a thick layer of fat below the skin
- large lungs and a large heart
- many small red blood cells
- presence of fetal haemoglobin throughout their adult life.

(a) Triglycerides are fats.

Describe the structure of a triglyceride.

(2)

- (b) Explain why large lungs and a large heart would help a yak to survive at high altitudes.

(3)

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- (c) The table shows some characteristics of the blood of yaks and the blood of humans.

Characteristic of blood	Blood of yaks	Blood of humans
Concentration of red blood cells/cells dm^{-3}	1.0×10^{13}	4.0×10^6 to 6.0×10^6
Volume of one red blood cell / fl	43	90
Concentration of haemoglobin/g dm^{-3}	136	12 to 18

- (i) The unit given for the volume of a red blood cell is a femtolitre (fl).

$$1 \text{ fl} = 1.0 \times 10^{-15} \text{ dm}^3$$

Give the volume of a red blood cell of a yak in cm^3 .

Express your answer in standard form.

(1)

Answer cm^3



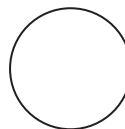
- (ii) The diagram gives some information about two spheres, sphere A and sphere B.

Sphere A



$\longleftrightarrow$
8 mm

Sphere B



$\longleftrightarrow$
16 mm

- sphere A has a surface area : volume ratio of 1 : 1.3
- sphere B has a surface area of 768 mm²

Calculate the surface area : volume ratio of sphere B.

Use the formula:

$$\text{Volume} = \frac{4}{3} \pi r^3$$

where $\pi = 3.0$

(2)

Answer

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



*(iii) Discuss why the blood of yaks enables these animals to live at high altitudes.

Use all the information in this question to support your answer.

(6)

(Total for Question 7 = 14 marks)

